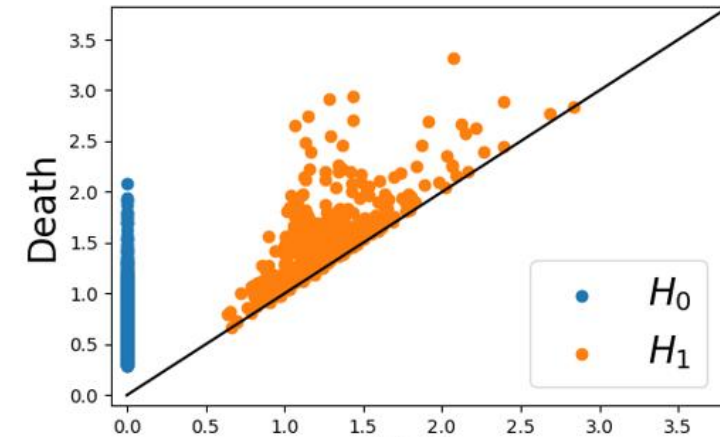
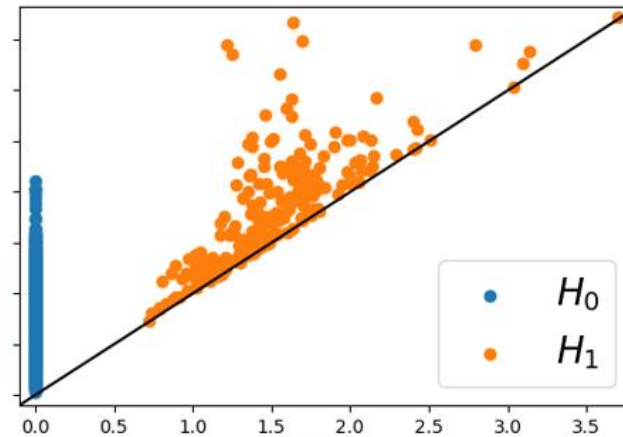
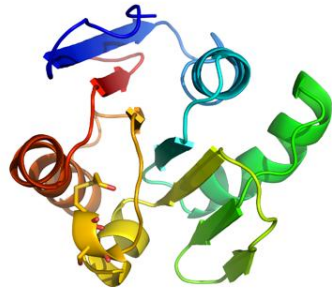
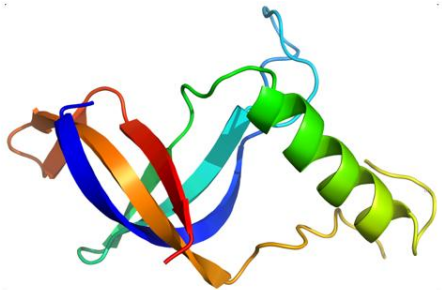




# EAFIT - TDA

Persistencia homológica y machine learning

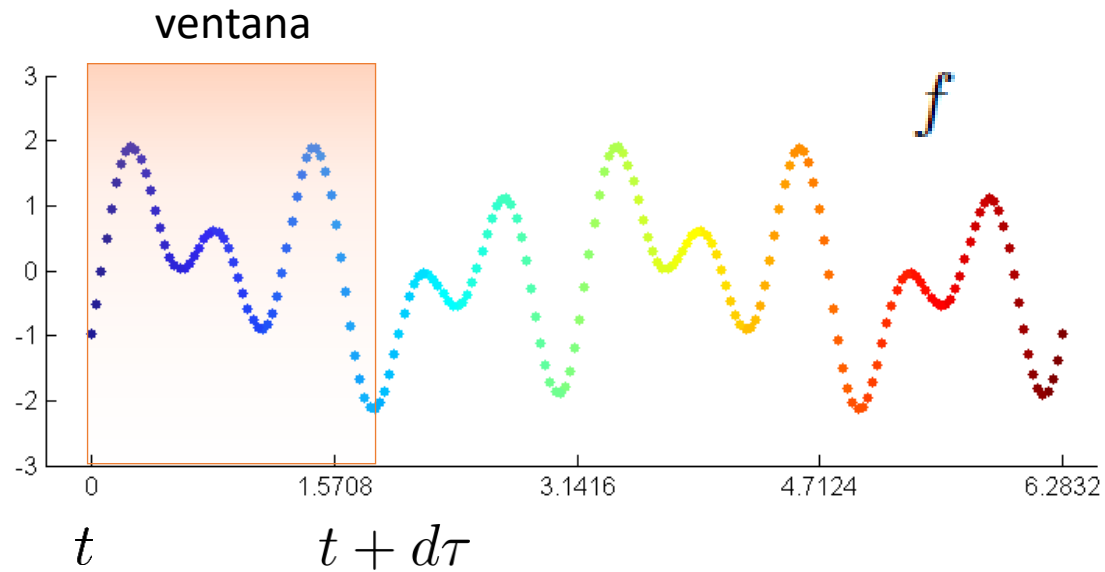




# A Comparative Study of Machine Learning Methods for Persistence Diagrams

*Danielle Barnes<sup>1\*</sup>, Luis Polanco<sup>1,2</sup> and Jose A. Perea<sup>1,2</sup>*

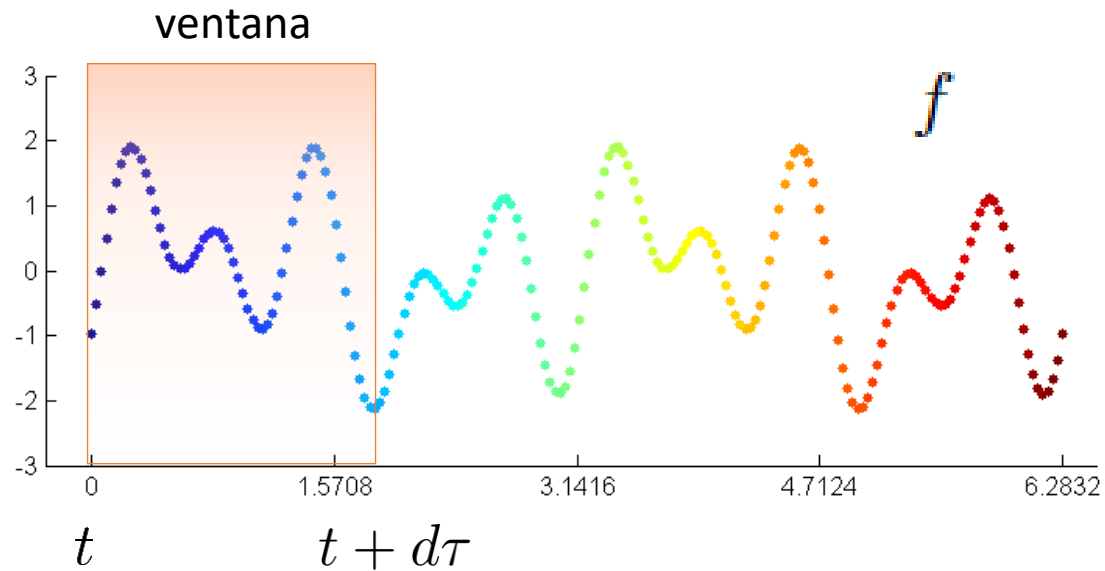
# Sliding window embedding (Encaje de ventanas deslizantes)



$\tau$  → Dilación

$d + 1$  → Dimensión

# Sliding window embedding (Encaje de ventanas deslizantes)

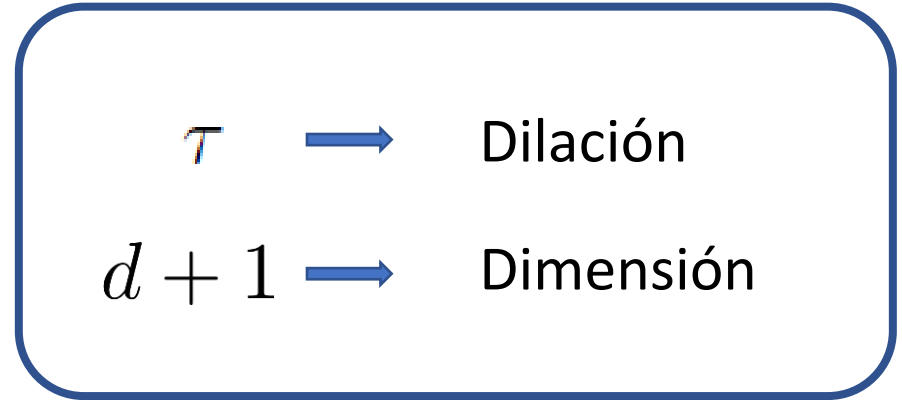
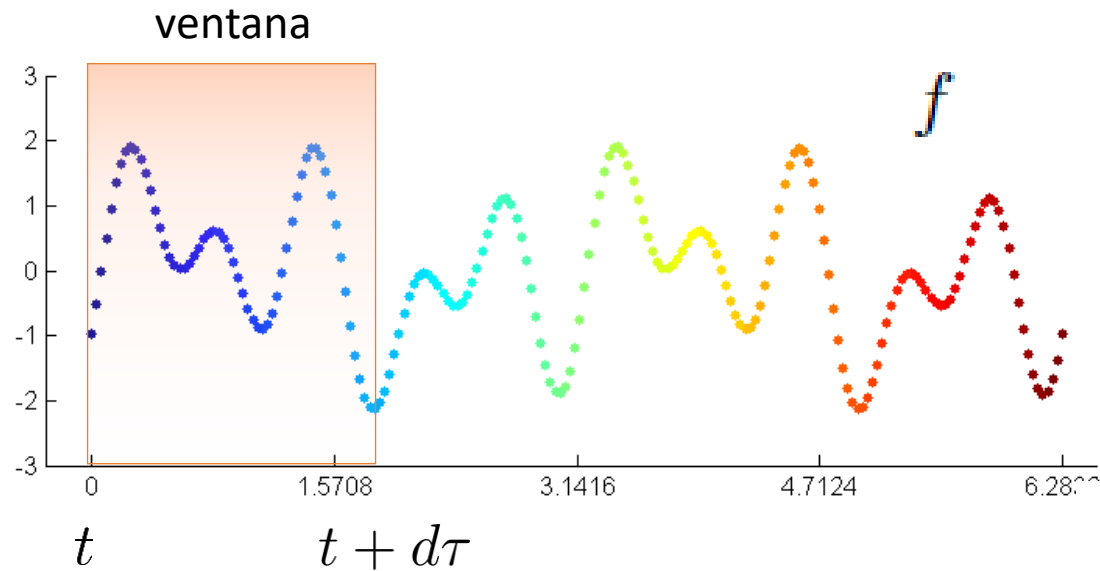


$\tau$  → Dilación

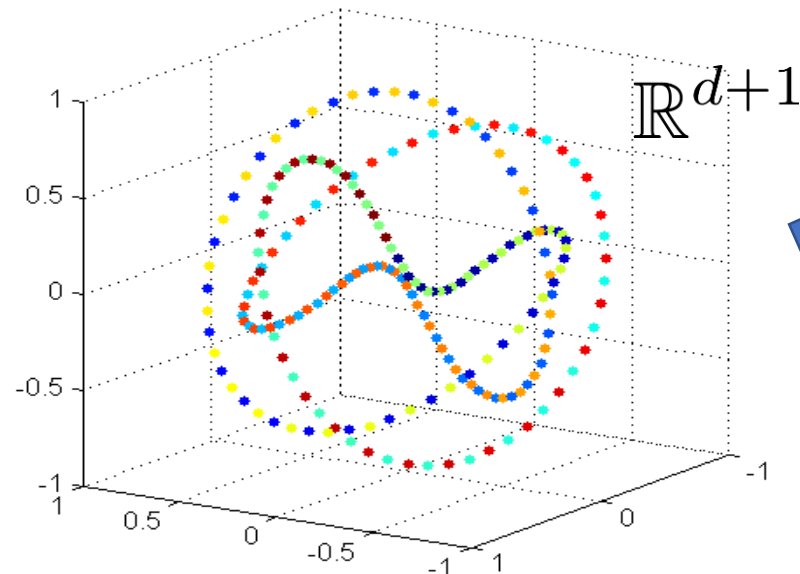
$d + 1$  → Dimensión

$$SW_{d,\tau}f(t) = \begin{bmatrix} f(t) \\ f(t + \tau) \\ \vdots \\ f(t + d\tau) \end{bmatrix}$$

# Sliding window embedding (Encaje de ventanas deslizantes)



$$SW_{d,\tau} f(t) = \begin{bmatrix} f(t) \\ f(t + \tau) \\ \vdots \\ f(t + d\tau) \end{bmatrix}$$

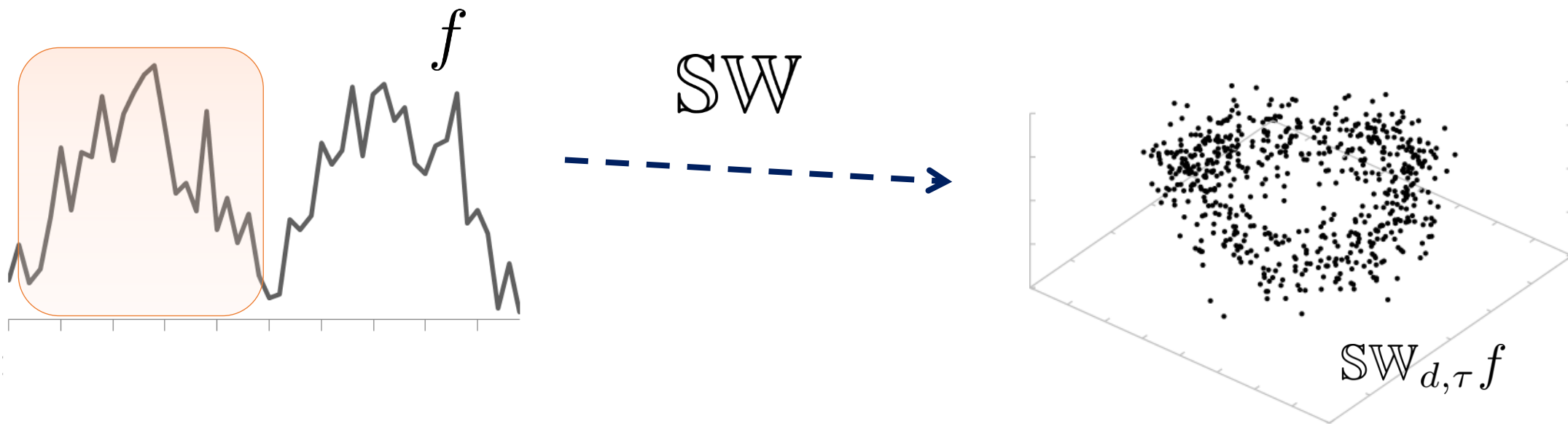


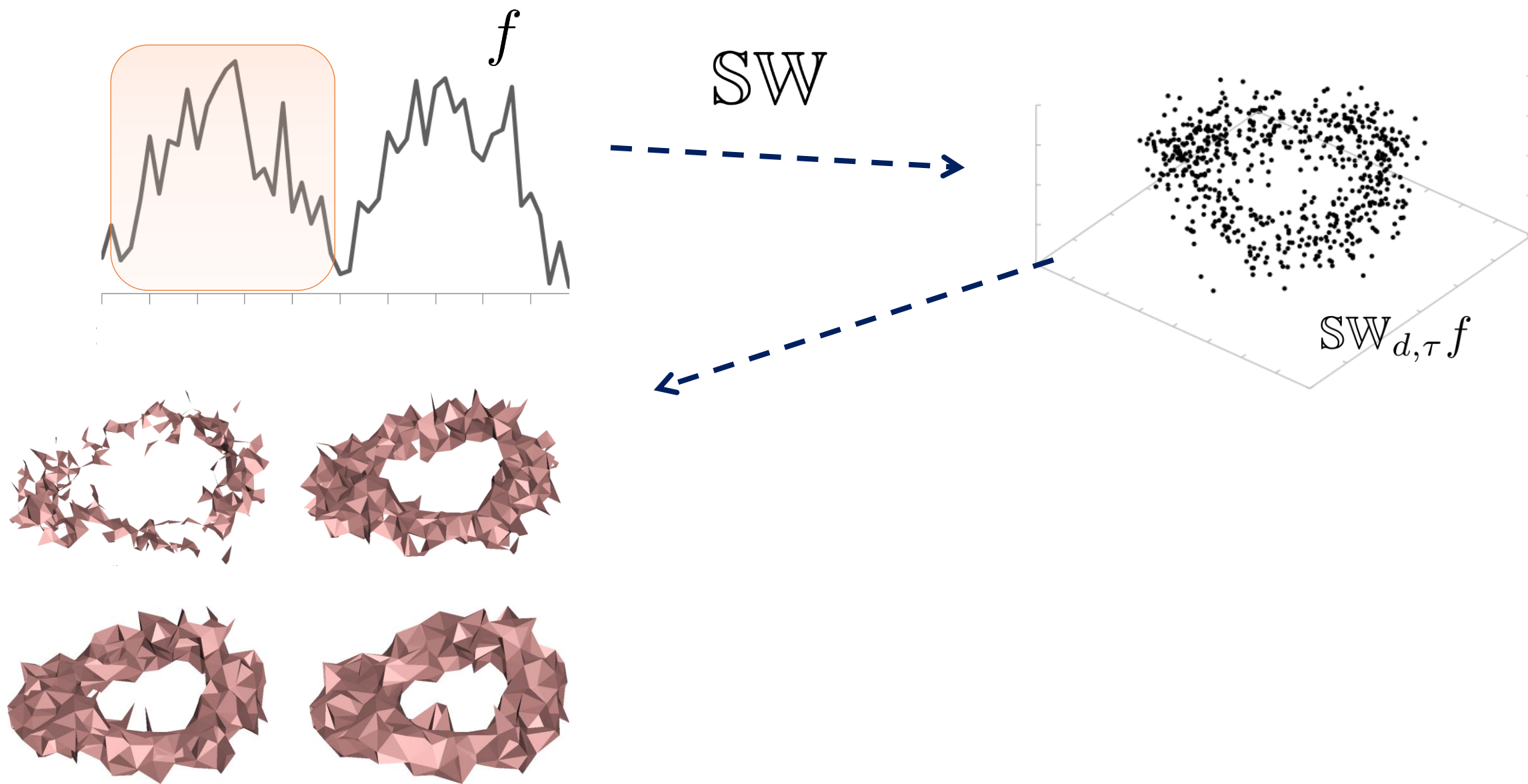
Nube de ventanas  
deslizantes

$$SW_{d,\tau} f \parallel SW_{d,\tau} f(I)$$

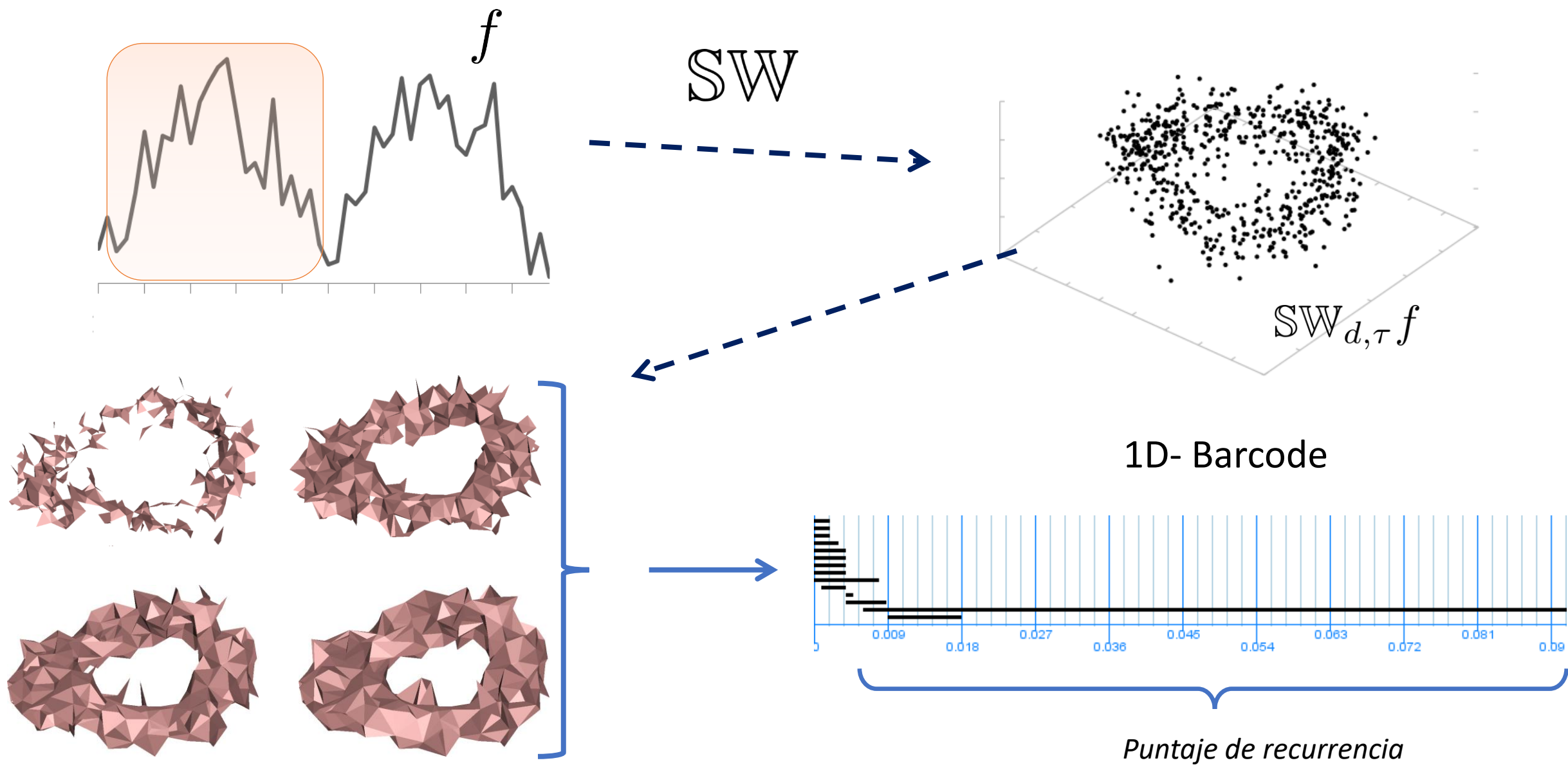
$$I \subset \mathbb{R}$$

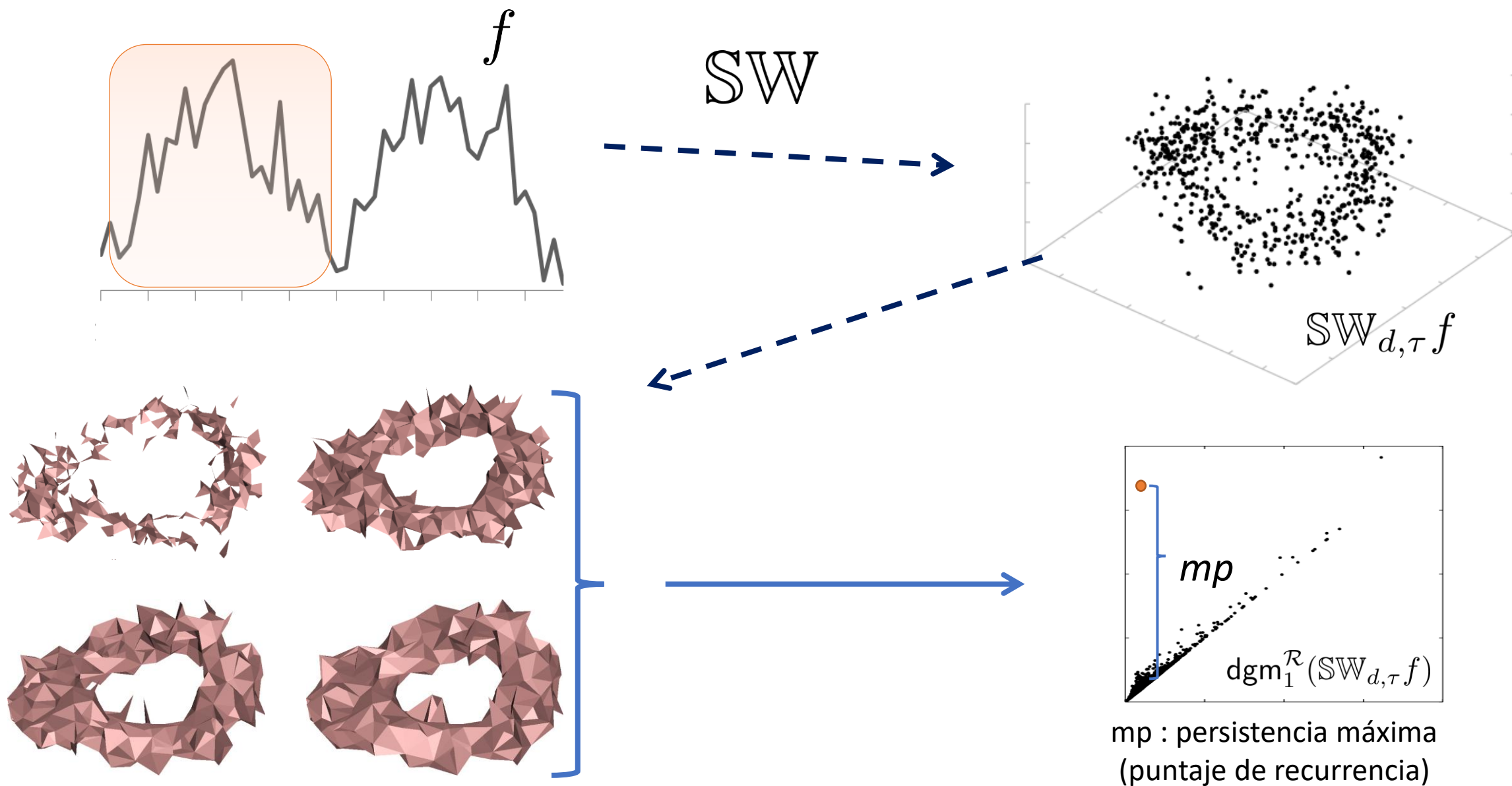
# SW1PerS: Sliding Windows and 1-Persistence Scoring



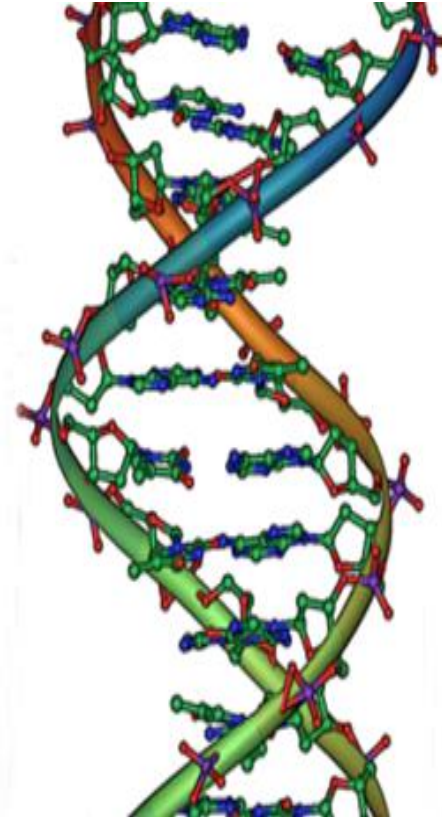








# Relojes Biológicos



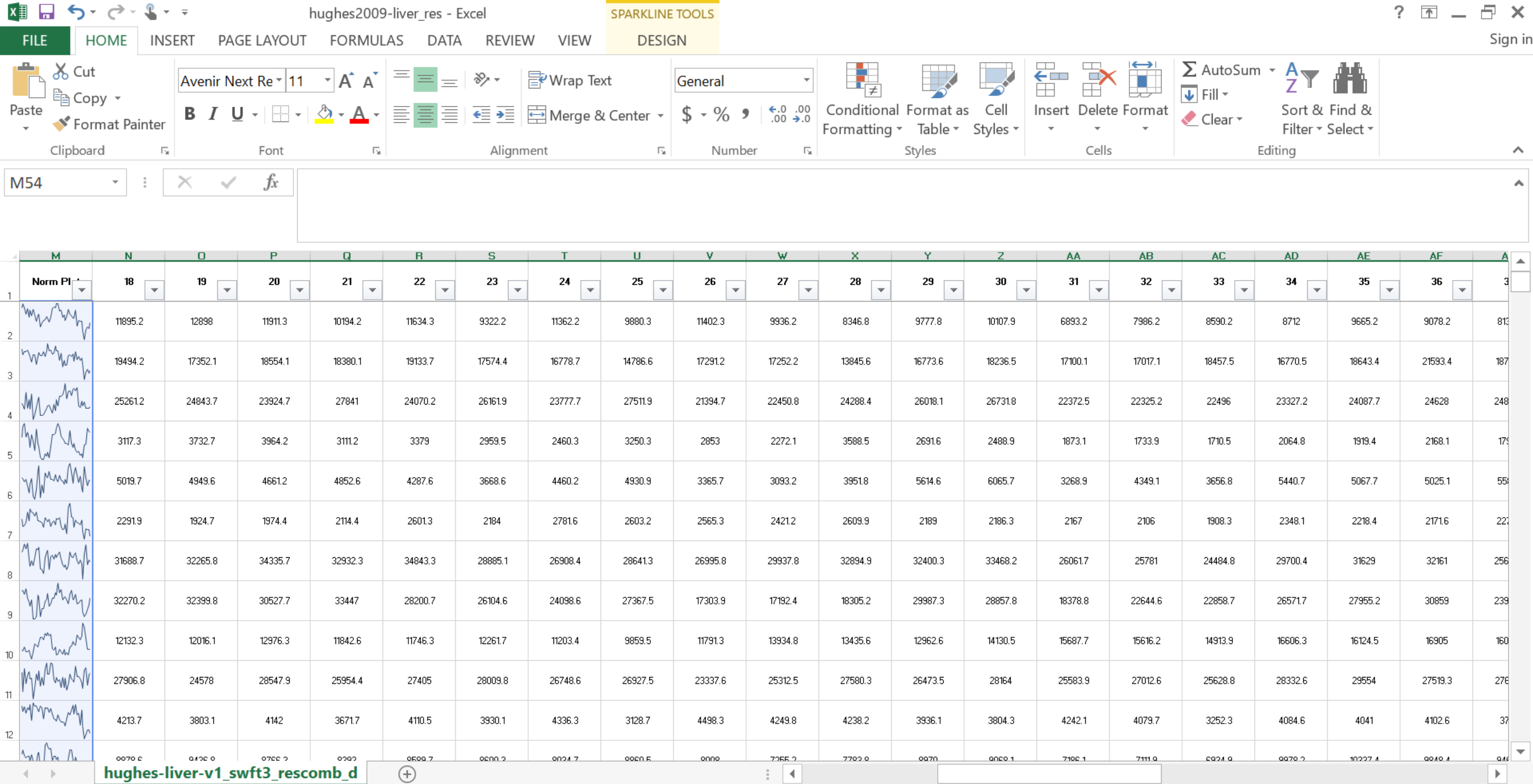
**METHODOLOGY ARTICLE**

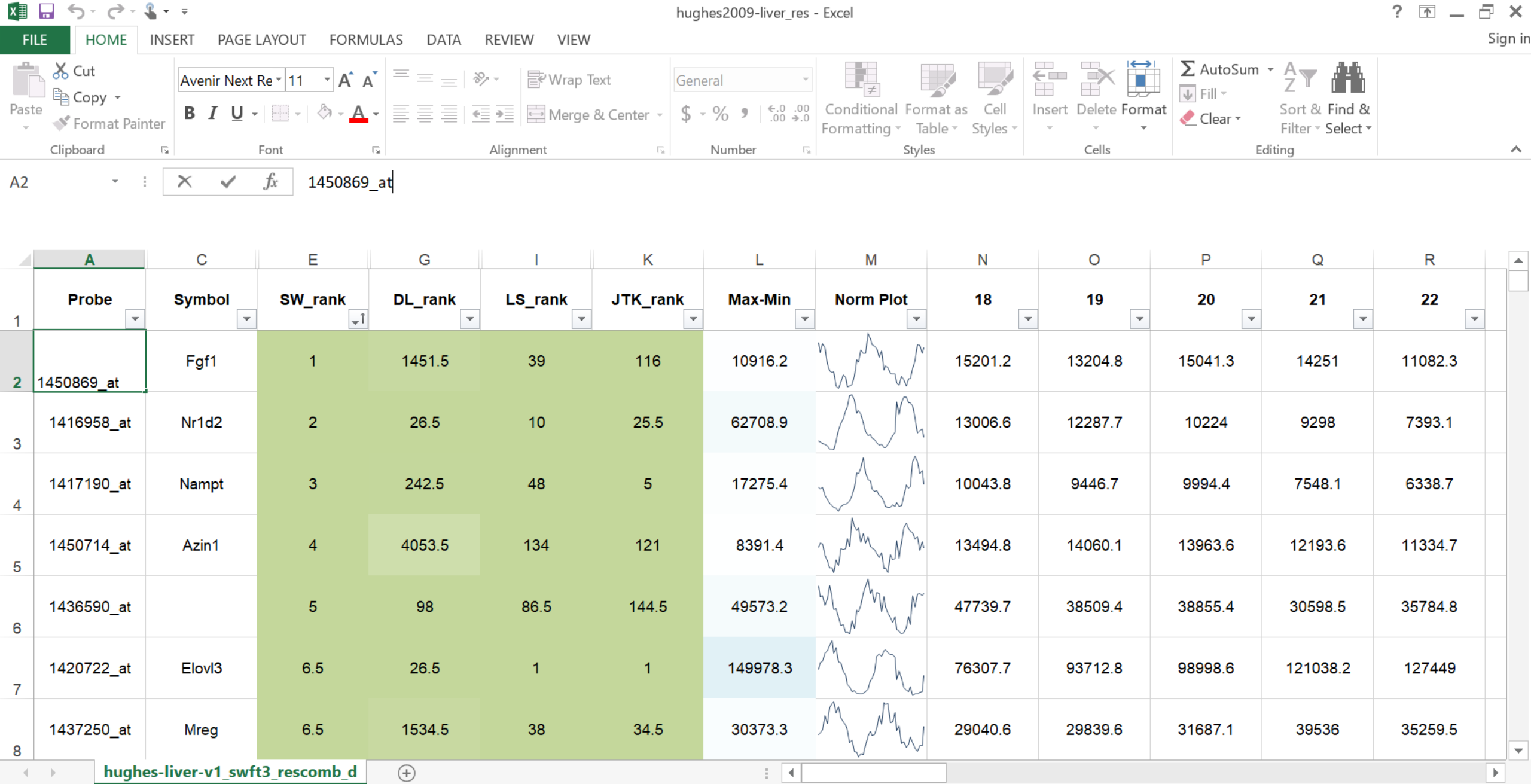
**Open Access**



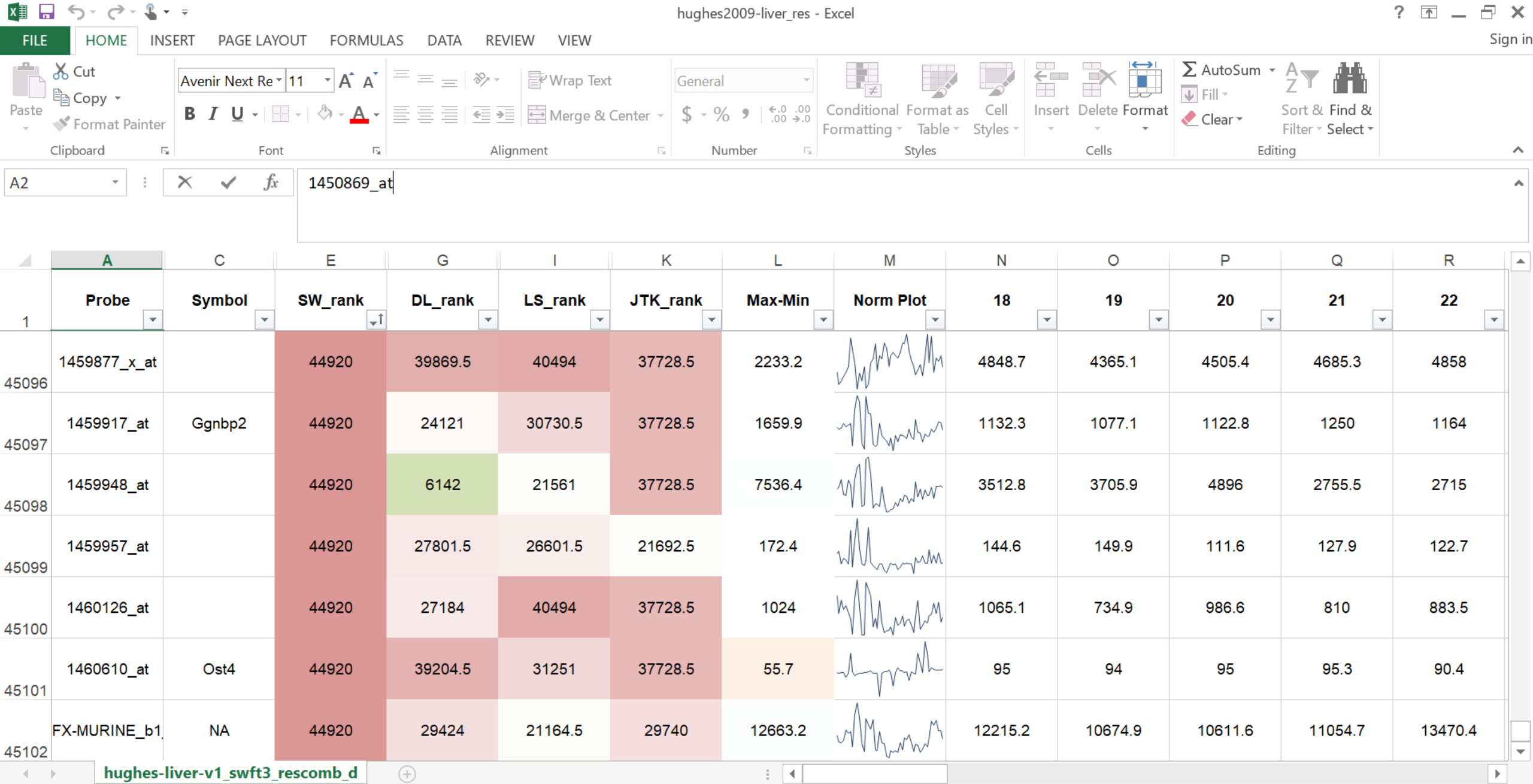
# SW1PerS: Sliding windows and 1-persistence scoring; discovering periodicity in gene expression time series data

Jose A. Perea<sup>1,2\*</sup>, Anastasia Deckard<sup>3</sup>, Steve B. Haase<sup>4,5</sup> and John Harer<sup>1,4,6</sup>

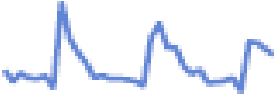








# Ciclo metabólico de la levadura

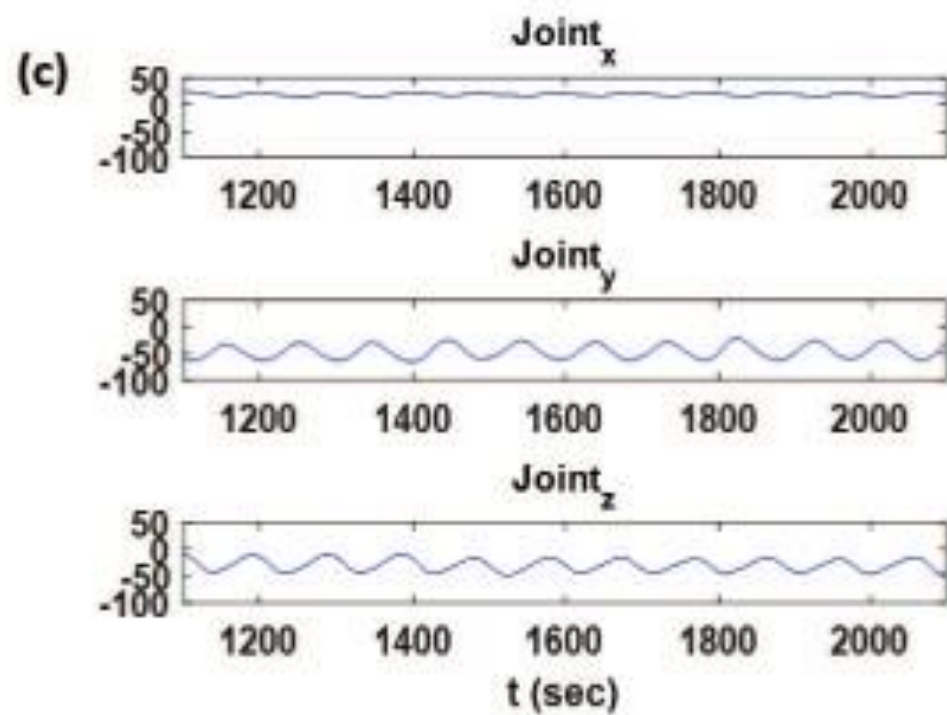
| Gene   | SW  | DL   | LS     | JTK    | Amp   | Plot                                                                                |
|--------|-----|------|--------|--------|-------|-------------------------------------------------------------------------------------|
| ECM33  | 137 | 1552 | 1194.5 | 1492   | 35.86 |  |
| CDC9   | 291 | 1494 | 1993.5 | 2714.5 | 2.81  |  |
| SAM1,2 | 628 | 1133 | 1723   | 3289.5 | 60.82 |  |
| MSH6   | 715 | 3569 | 2381   | 3341.5 | 5.06  |  |

Rankings de genes en el top 10% (total 9,330 ) de acuerdo a SW, y no en el top 10% de ningún otro algoritmo

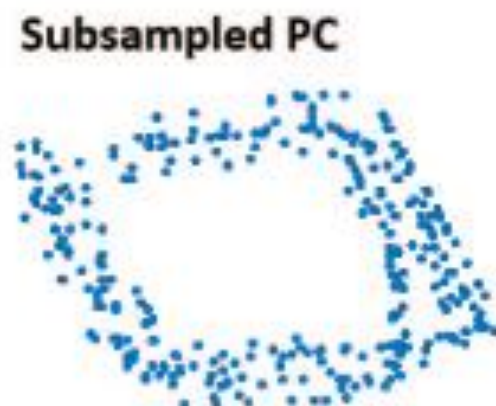


# **ACTION CLASSIFICATION FROM MOTION CAPTURE DATA USING TOPOLOGICAL DATA ANALYSIS**

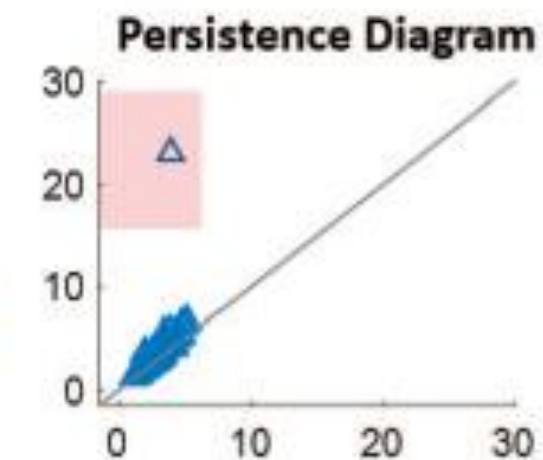
*Alireza Dirafzoon, Namita Lokare and Edgar Lobaton*



(d)

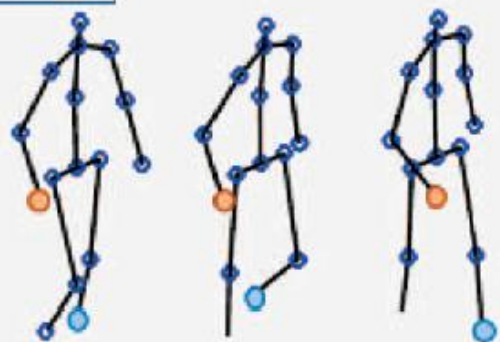


(e)

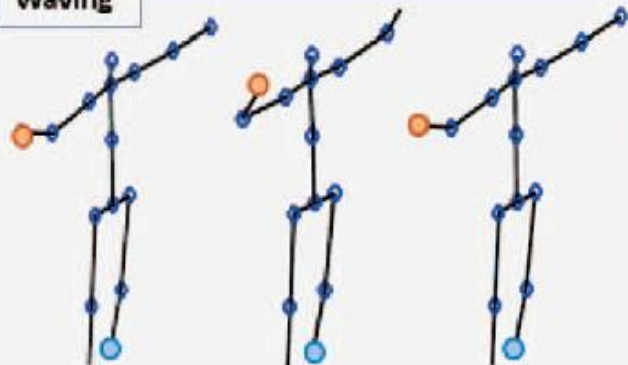


(f)

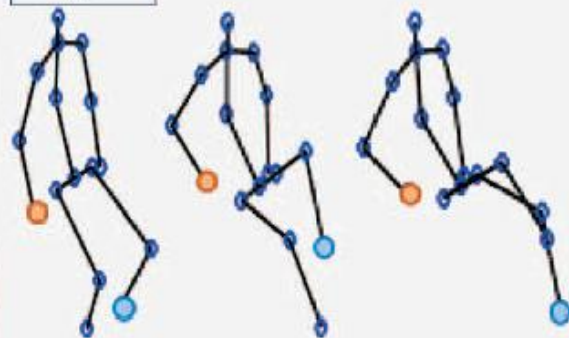
Walking



Waving



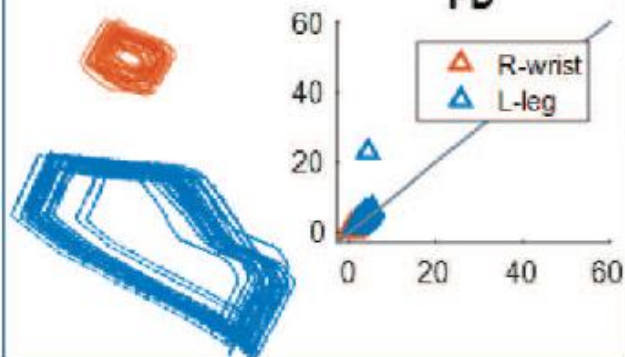
Bicycle



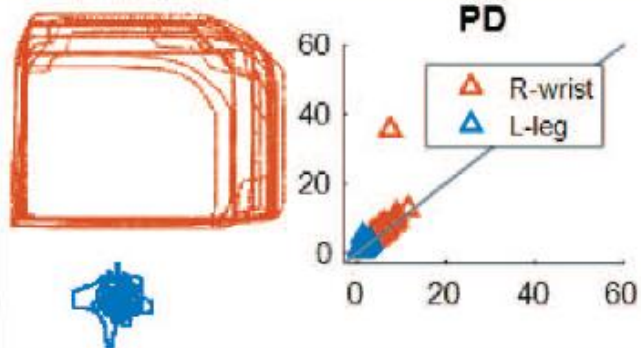
Golfing



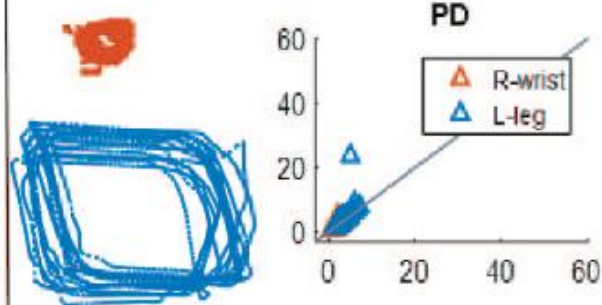
PD



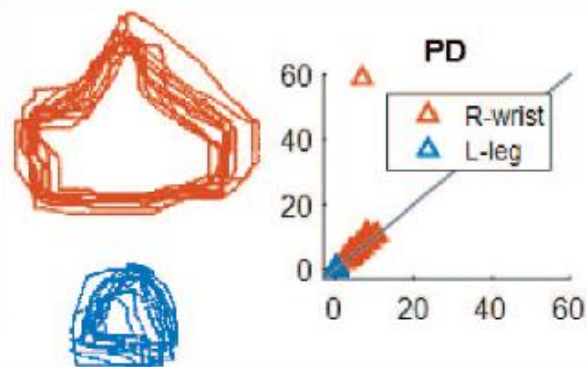
PD

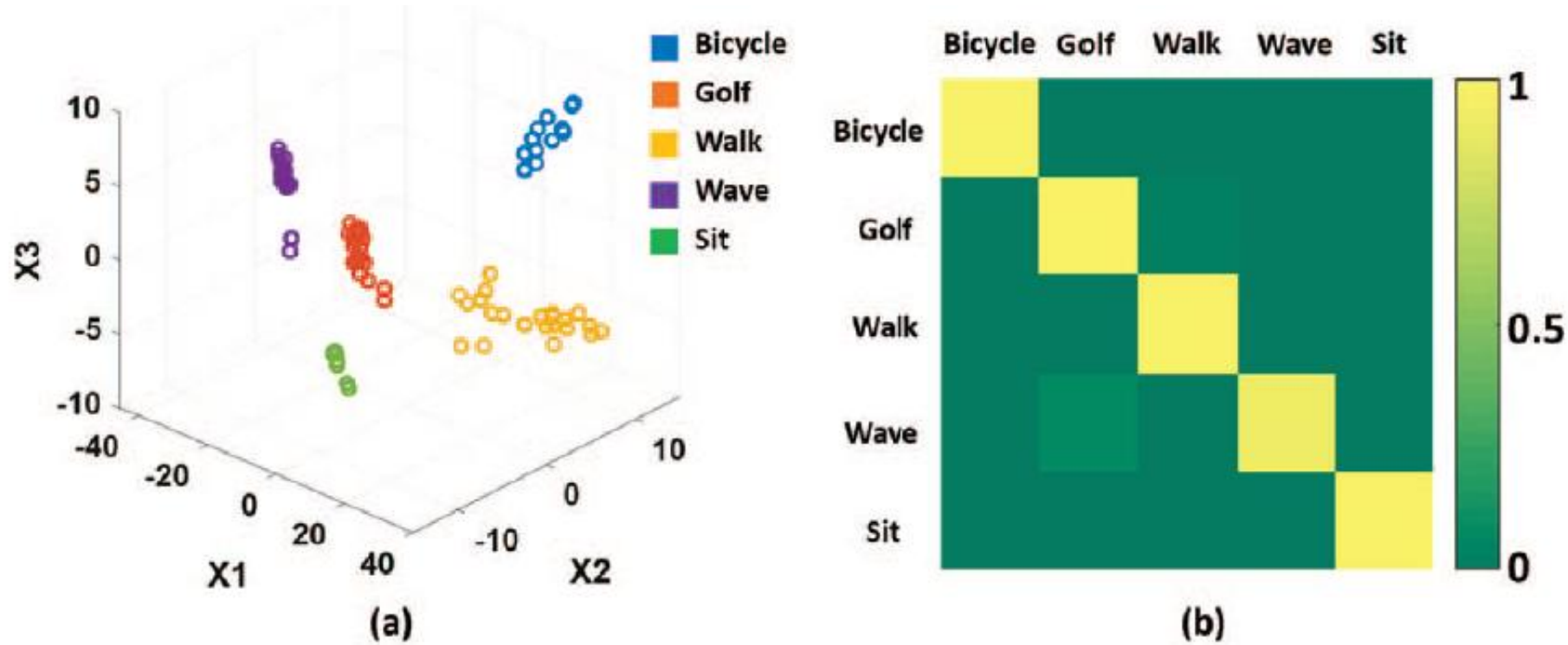


PD



PD





**Fig. 4.** (a) Separation of classes from the training set, (b) Confusion matrix over the predicted and true classes

**Table 1.** Class accuracy results for the activities

| Action   | Bicycle | Golf   | Walk   | Wave   | Sit  |
|----------|---------|--------|--------|--------|------|
| Accuracy | 1.00    | 0.9787 | 0.9929 | 0.9858 | 1.00 |



RESEARCH ARTICLE **OPEN ACCESS**

# Topological Data Analysis Based Characteristics of Electroencephalogram Signals in Children With Sleep Apnea

Aarti Sathyanarayana<sup>1,2</sup> | Shashank Manjunath<sup>2</sup>  | Jose A. Perea<sup>2,3</sup>

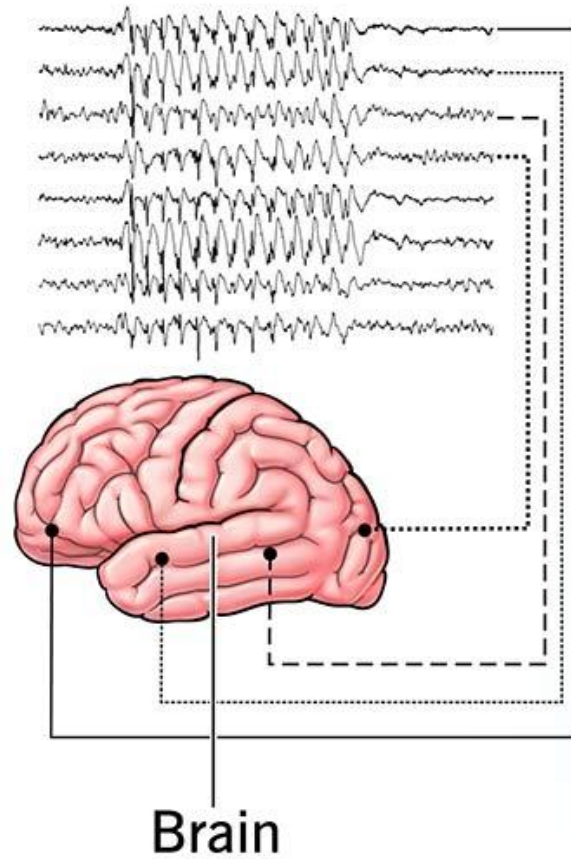
<sup>1</sup>Bouvé College of Health Sciences, Northeastern University, Boston, Massachusetts, USA | <sup>2</sup>Khoury College of Computer Sciences, Northeastern University, Boston, Massachusetts, USA | <sup>3</sup>Department of Mathematics, Northeastern University, Boston, Massachusetts, USA

**Correspondence:** Aarti Sathyanarayana ([a.sathyanarayana@northeastern.edu](mailto:a.sathyanarayana@northeastern.edu))

**Received:** 10 October 2024 | **Revised:** 4 February 2025 | **Accepted:** 5 February 2025

# Electroencephalogram (EEG)

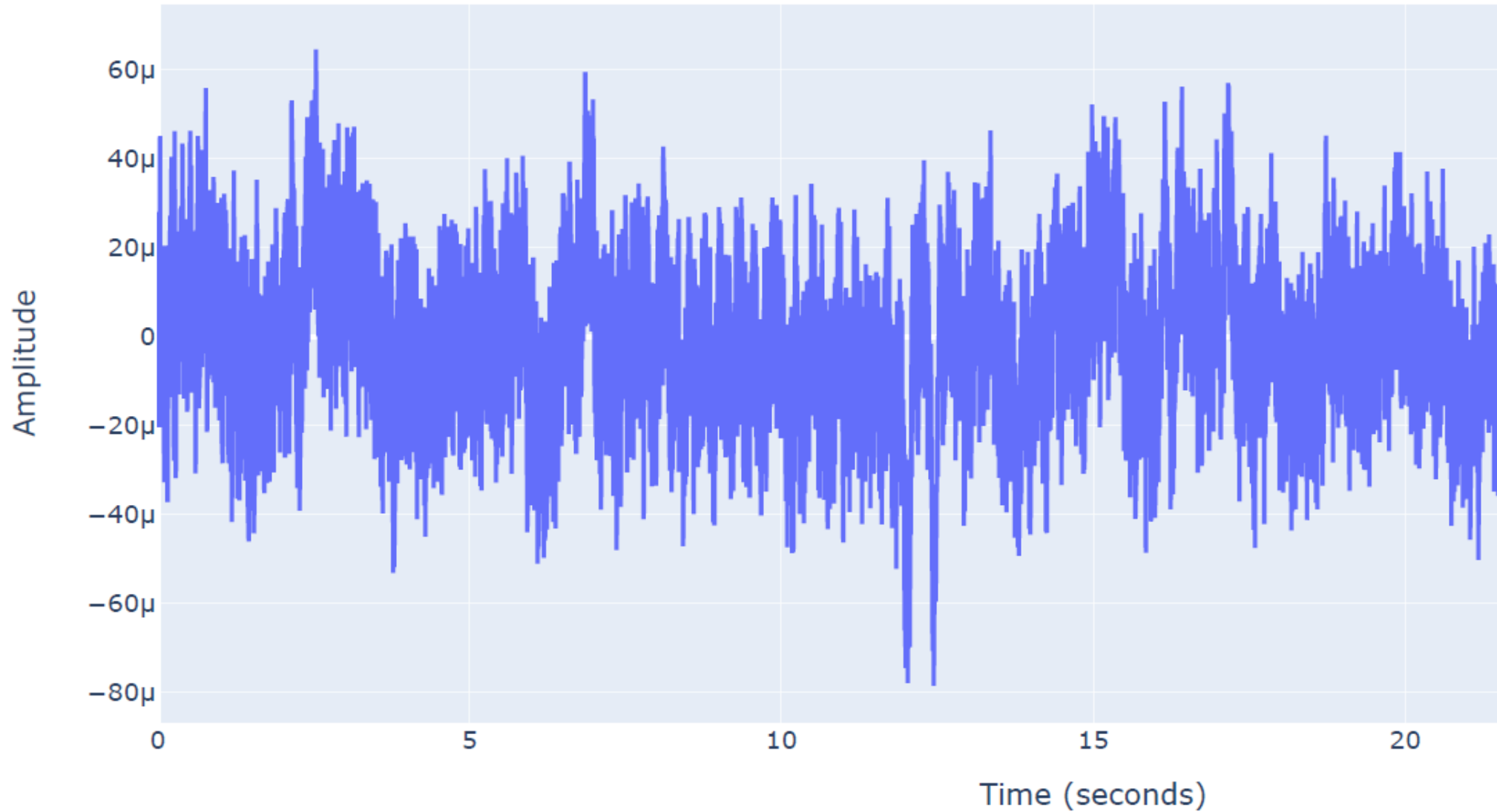
EEG (scan of brainwaves)



Electrodes  
glued to scalp



## EEG Signal for F3-M2



# Datos:

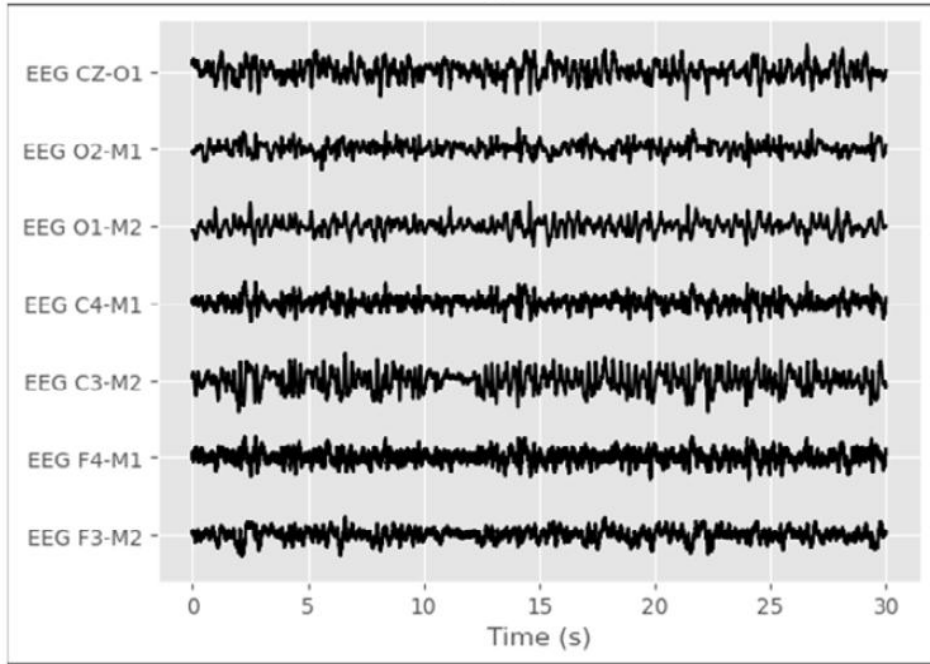
## *Nationwide Children's Hospital Sleep DataBank (NCHSDB)*

- 3.984 estudios de sueño
- 3.673 pacientes únicos

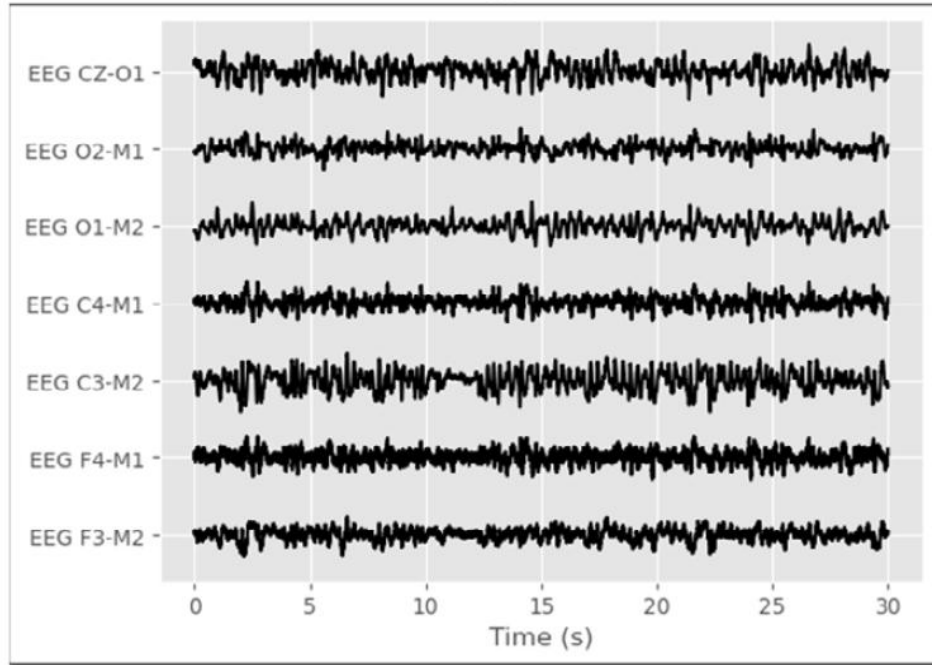




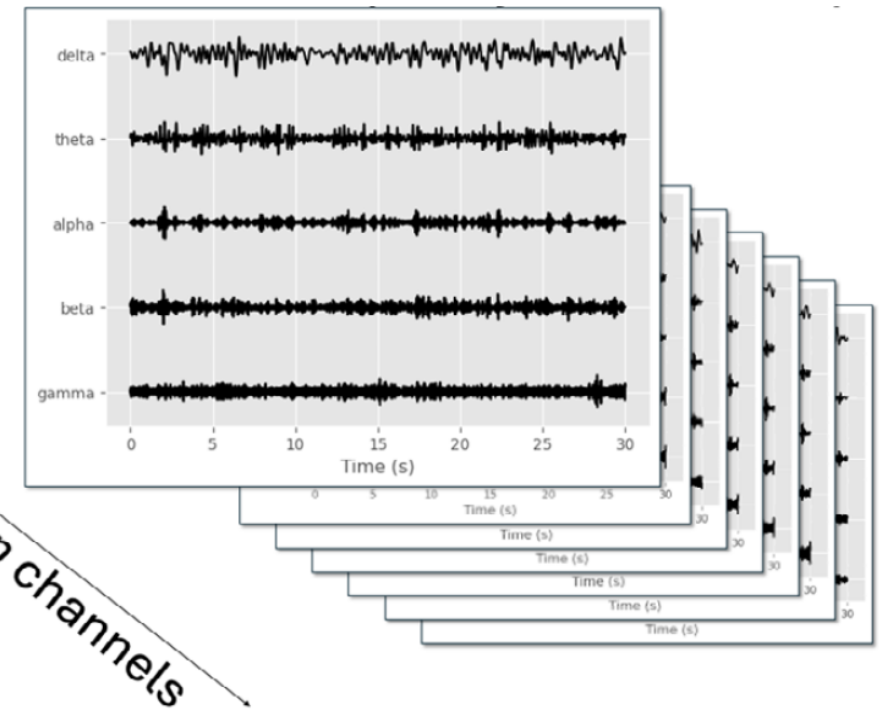
## Raw Signals



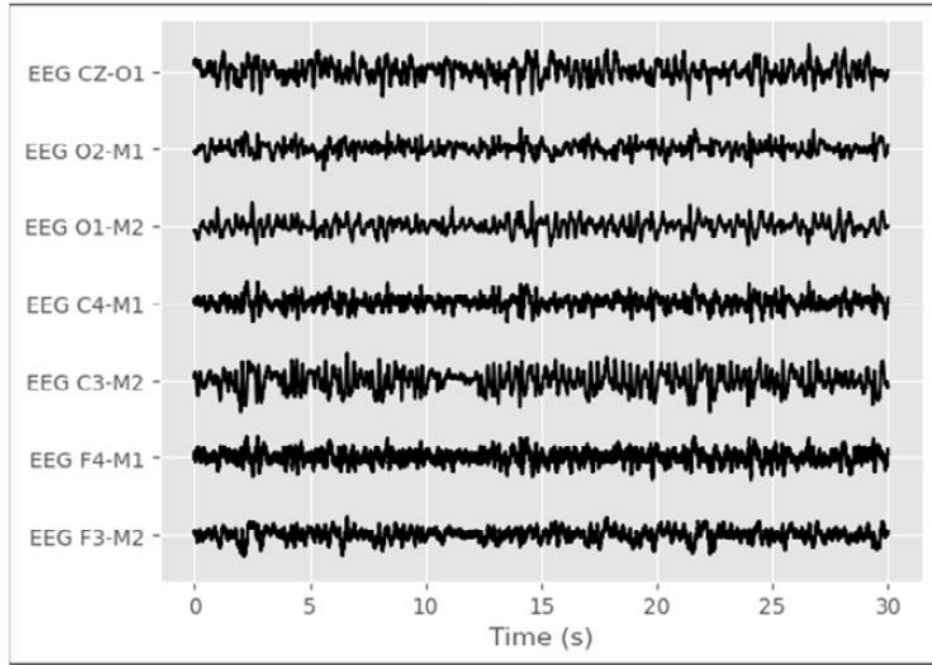
## Raw Signals



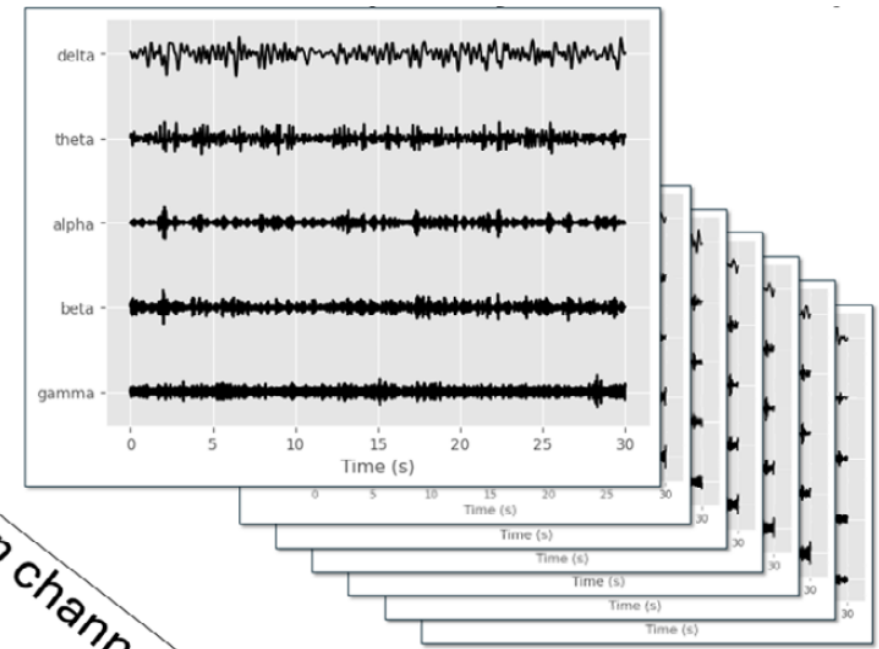
## Fourier (EEG Bands)



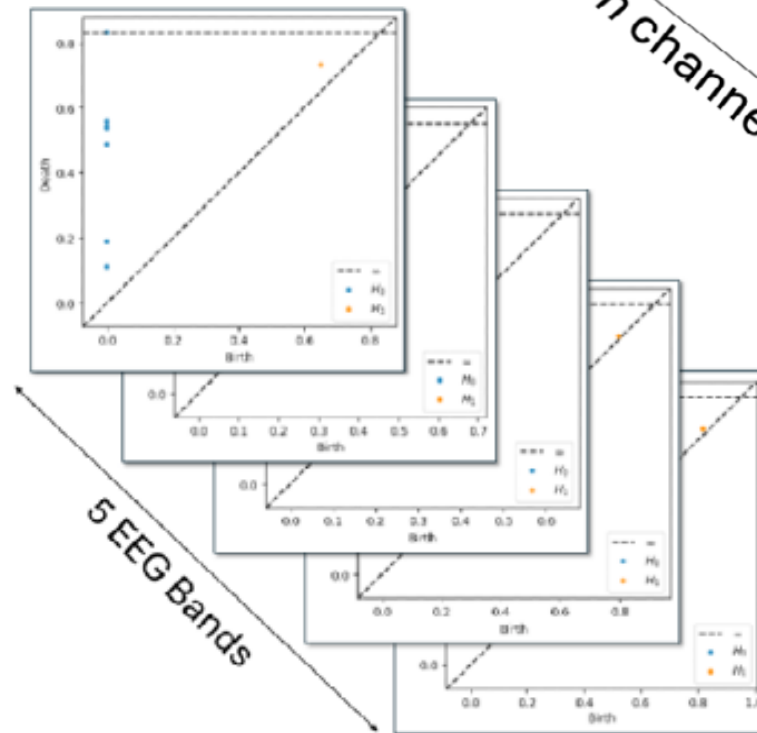
## Raw Signals



## Fourier (EEG Bands)



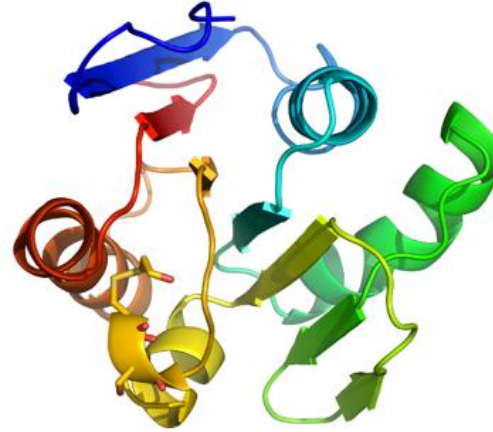
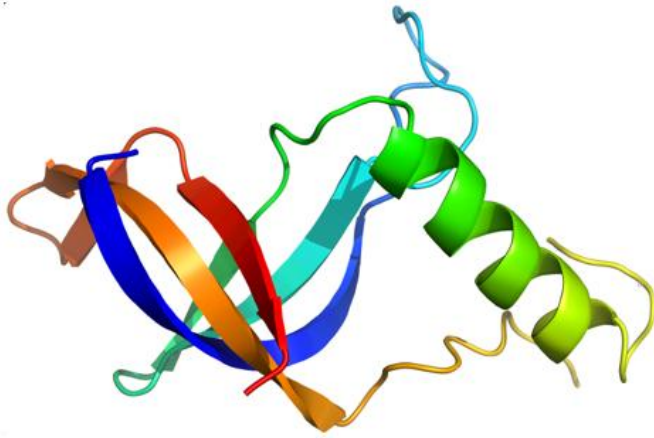
$n$  channels



5 EEG Bands

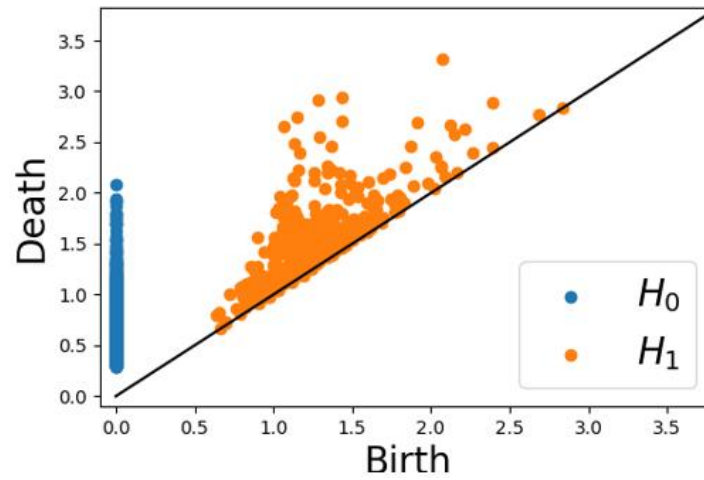
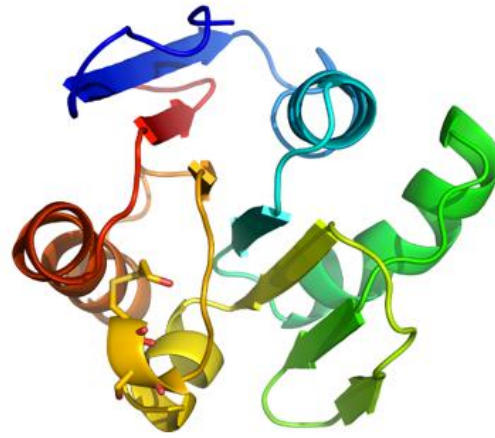
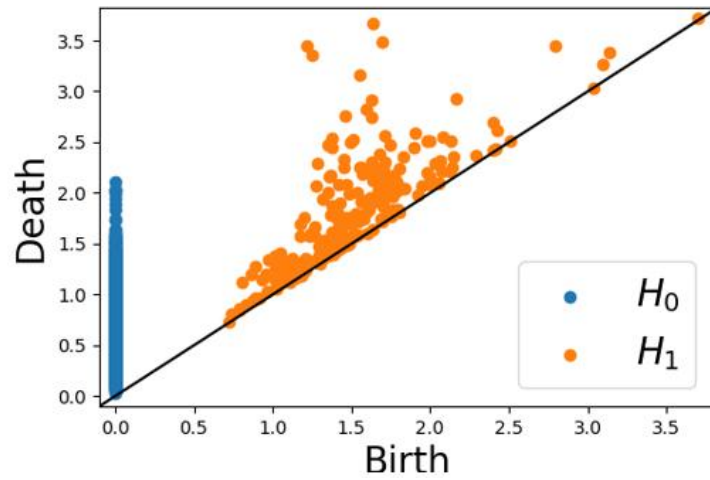
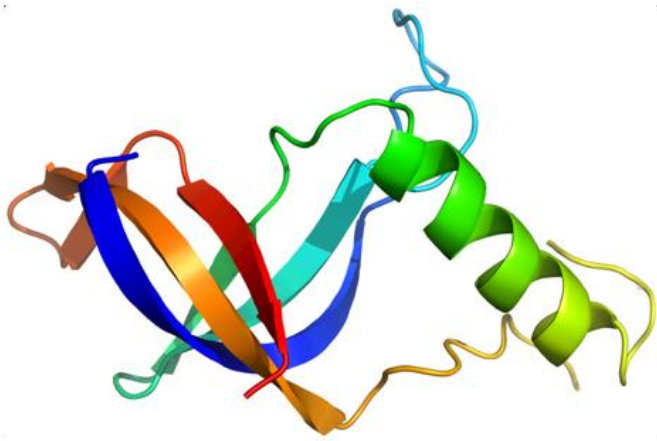
**TDA**

|                   |       | NREM1<br>(276 mild/1054 normal)     | NREM2<br>(656 mild/1063 normal)     | NREM3<br>(310 mild/1059 normal)     | REM<br>(1023 mild/1050 normal)    |
|-------------------|-------|-------------------------------------|-------------------------------------|-------------------------------------|-----------------------------------|
| Apnea<br>Leve     | Delta | 0.864                               | 0.003                               | 0.559                               | 0.019                             |
|                   | Theta | 0.967                               | 0.225                               | 0.770                               | 0.346                             |
|                   | Alpha | 0.047                               | 0.006                               | 0.577                               | 0.949                             |
|                   | Beta  | 0.001                               | 0.001                               | 0.001                               | 0.381                             |
|                   | Gamma | 0.001                               | 0.001                               | 0.001                               | 0.040                             |
| Apnea<br>Moderada |       | NREM1<br>(142 moderate/1054 normal) | NREM2<br>(268 moderate/1063 normal) | NREM3<br>(152 moderate/1059 normal) | REM<br>(342 moderate/1050 normal) |
|                   | Delta | 0.124                               | 0.005                               | 0.630                               | 0.415                             |
|                   | Theta | 0.996                               | 0.088                               | 0.951                               | 0.728                             |
|                   | Alpha | 0.519                               | 0.003                               | 0.820                               | 0.974                             |
|                   | Beta  | 0.014                               | 0.009                               | 0.686                               | 0.809                             |
|                   | Gamma | 0.048                               | 0.001                               | 0.002                               | 0.008                             |
| Apnea<br>Severa   |       | NREM1<br>(340 severe/1054 normal)   | NREM2<br>(451 severe/1063 normal)   | NREM3<br>(310 severe/1059 normal)   | REM<br>(464 severe/1050 normal)   |
|                   | Delta | 0.927                               | 0.937                               | 1.000                               | 0.106                             |
|                   | Theta | 0.016                               | 0.738                               | 1.000                               | 0.921                             |
|                   | Alpha | 0.069                               | 0.453                               | 0.831                               | 1.000                             |
|                   | Beta  | 0.001                               | 0.048                               | 0.502                               | 0.005                             |
|                   | Gamma | 0.002                               | 0.001                               | 0.001                               | 0.001                             |

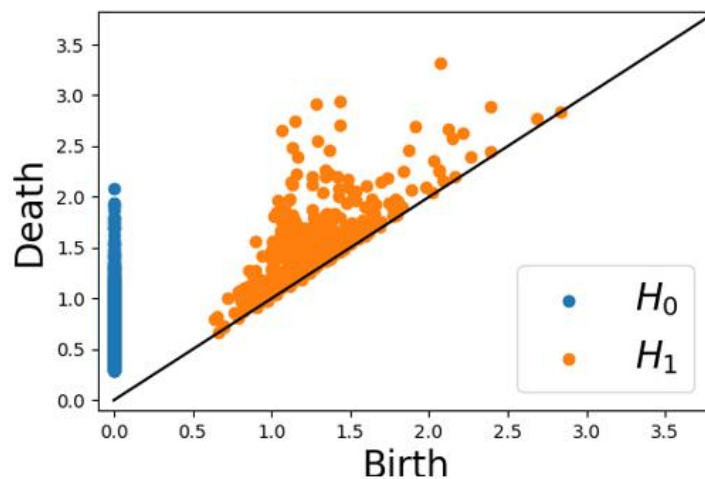
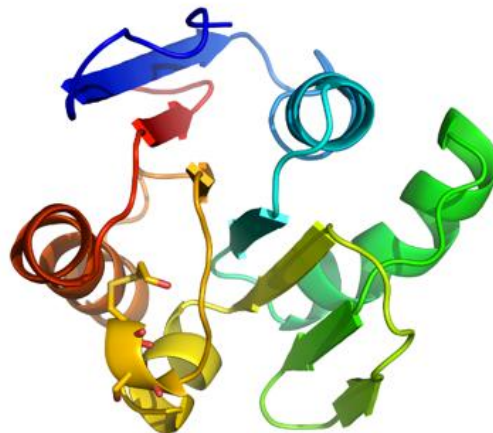
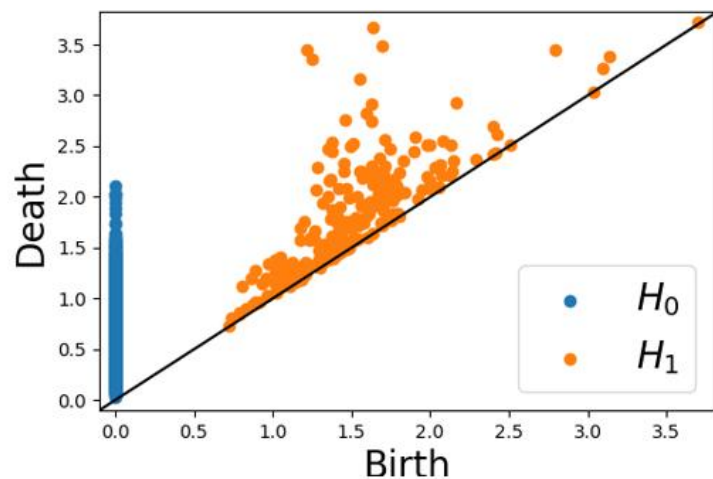
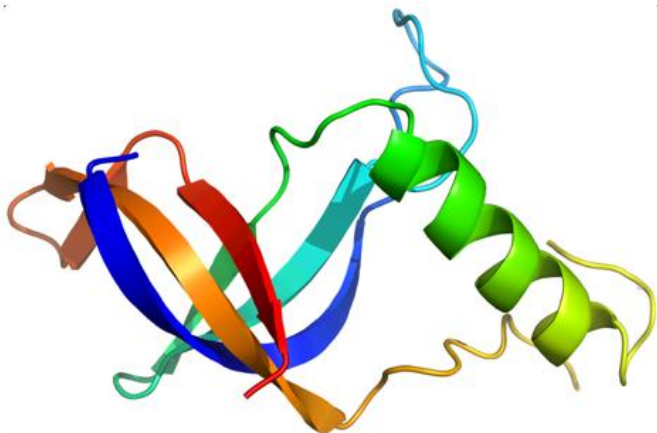


## Protein Classification Benchmark Collection (PCB00019) – SCOP40mini

|                |              |                         |
|----------------|--------------|-------------------------|
| 1,357 proteins | # atoms ~ 1K | 55 classification tasks |
|----------------|--------------|-------------------------|







|                             |            | Train           | Test                  |
|-----------------------------|------------|-----------------|-----------------------|
| CDER                        | Polynomial | $0.90 \pm 0.07$ | $0.98 \pm 0.02$       |
|                             | RBF        | $0.91 \pm 0.06$ | $0.97 \pm 0.02$       |
|                             | Sigmoid    | $0.90 \pm 0.07$ | $0.98 \pm 0.02$       |
| Topological features in [4] |            | -               | $0.82 \pm \text{---}$ |